

QVerse — Week 8: Measurement, Shots, and Statistical Thinking

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `08_measurement_shots_statistics.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 8; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Interpret finite-shot counts as samples.
- Compare statevector probabilities with empirical frequencies.
- Measure in different bases.
- Observe approximate $1/\sqrt{N}$ sampling-error scaling.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. Measurement produces samples

A quantum circuit defines a probability distribution over measurement outcomes. A finite execution returns samples from that distribution.

If the theoretical probability of outcome 1 is p , then one shot gives a Bernoulli random variable

$$X = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

After N independent shots, the number of observed 1s is binomial:

$$K \sim \text{Binomial}(N, p).$$

The empirical estimate is

$$\hat{p} = \frac{K}{N}.$$

2. Finite-shot fluctuations

The expected value of the empirical estimator is

$$\mathbb{E}[\hat{p}] = p.$$

Its variance is

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{N},$$

so the standard error is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{N}}.$$

The characteristic scaling is therefore

$$\sigma_{\hat{p}} \propto \frac{1}{\sqrt{N}}.$$

To reduce sampling error by a factor of 10, roughly 100 times more shots are needed.

This $1/\sqrt{N}$ law is a key experimental fact.

3. Example: a 50/50 state

For $p = 1/2$,

$$\sigma_{\hat{p}} = \frac{1}{2\sqrt{N}}.$$

At $N = 100$,

$$\sigma_{\hat{p}} = 0.05.$$

At $N = 10,000$,

$$\sigma_{\hat{p}} = 0.005.$$

Therefore a 50/50 quantum state should not be expected to return exactly equal counts at finite N .

Two correct runs may produce different histograms because the samples are different.

4. Statevector probability versus empirical frequency

For a state

$$|\psi\rangle = \sum_x \alpha_x |x\rangle,$$

the theoretical probability is

$$p_x = |\alpha_x|^2.$$

The observed frequency is

$$f_x = \frac{n_x}{N},$$

where n_x is the number of occurrences of x .

As N increases, f_x tends to approach p_x , but finite discrepancies are normal.

This distinction is essential when comparing simulator statevectors, shot-based simulation, and hardware data.

5. Measurement basis

The measurement outcome depends on the basis.

Computational-basis or Z-basis measurement distinguishes

$$|0\rangle, \quad |1\rangle.$$

X-basis measurement distinguishes

$$|+\rangle, \quad |-\rangle.$$

To implement an X-basis measurement using ordinary computational readout, apply H first.

Thus:

$$\text{X-basis measurement} = H + \text{Z-basis measurement}.$$

For $|+\rangle$:

- Z-basis measurement is 50/50;
- X-basis measurement is deterministic $+$.

For $|-\rangle$:

- Z-basis measurement is 50/50;
- X-basis measurement is deterministic $-$.

6. Observables and eigenvalues

A measurement can also be described using an **observable**, represented by a Hermitian operator.

For Pauli Z ,

$$Z|0\rangle = +|0\rangle, \quad Z|1\rangle = -|1\rangle.$$

So a Z measurement can be associated with outcomes $+1$ and -1 .

The expectation value is

$$\langle Z \rangle = (+1)P(0) + (-1)P(1) = P(0) - P(1).$$

For $|+\rangle$,

$$P(0) = P(1) = \frac{1}{2},$$

so

$$\langle Z \rangle = 0.$$

This expectation-value viewpoint becomes central in Week 13.

7. Statistical uncertainty is not hardware noise

Finite-shot randomness occurs even in a perfect noiseless quantum experiment.

Hardware noise adds additional imperfections such as:

- incorrect gates;
- readout errors;
- decoherence.

These are separate sources of deviation.

Increasing shots reduces sampling uncertainty but does not automatically remove hardware bias.

This distinction is vital for interpreting real experiments.

8. Relative error and rare events

For small probabilities, absolute and relative error behave differently.

If $p = 0.1$,

$$\sigma_{\hat{p}} = \sqrt{\frac{0.1(0.9)}{N}}.$$

If p is extremely small, many shots may be required to observe the event reliably at all.

Therefore shot count should be chosen based on the precision needed for the quantity of interest, not by using the same arbitrary number for every experiment.

9. Confidence intervals: intuitive view

A rough large-sample confidence interval can be written as

$$\hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}},$$

where $z \approx 1.96$ for an approximate 95% interval.

For beginner work, the goal is not to master statistical inference. The important habit is to report enough information that readers can judge uncertainty:

- shot count;
- observed counts or frequencies;
- expected ideal probability;
- a simple uncertainty estimate when useful.

10. Repeated experiments and random seeds

A simulator may use pseudorandom sampling. Fixing a random seed can make a demonstration reproducible.

However, reproducibility should not be confused with the physical statement that counts must always be identical. The theoretical object is the underlying distribution.

A good report can show both:

- a fixed-seed run for reproducible code;
- the theoretical probability that explains why other correct runs may differ.

11. Measurement changes accessible information

A single measurement basis generally does not reveal the full quantum state.

For one qubit, reconstructing an unknown state requires information from multiple complementary measurements. Full reconstruction is called **quantum state tomography**.

Tomography is beyond the notebook, but it reinforces a central point:

a histogram in one basis is only a partial view of the state.

12. Common misconceptions

“50/50 means exactly half the shots must be each outcome”

No. 50/50 is a probability statement.

“If two executions differ, one must be wrong”

Finite random samples naturally differ.

“More shots remove all errors”

They reduce sampling uncertainty, not systematic hardware errors.

“Measurement basis is just a plotting choice”

No. Changing the basis changes which physical observable is measured.

13. What to know before the notebook

You should be able to explain:

$$\hat{p} = \frac{K}{N},$$
$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{N}},$$

the $1/\sqrt{N}$ scaling, and the difference between Z-basis and X-basis measurement.

You should also understand why finite-shot variation is expected even for a perfectly correct circuit.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.